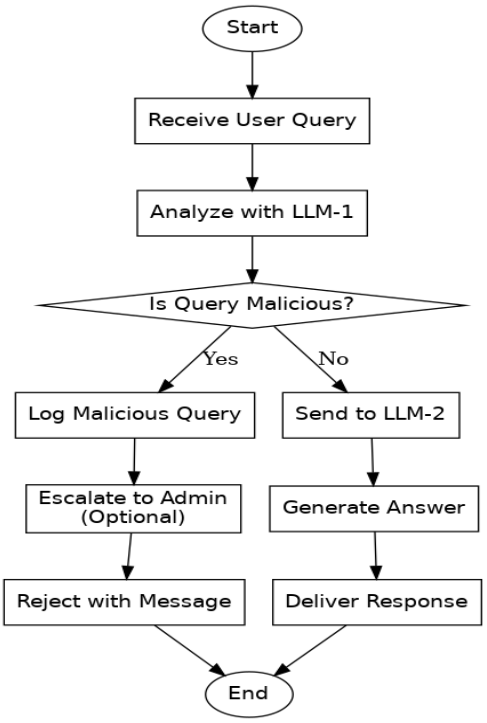


FIGURE 7. Technical schematic of the dual-LLM architecture (with LLM-1 implementing the Intent Identifier and Deflector functions.

Performance analysis reveals that the dual-LLM architecture introduces an 18% latency overhead compared with single-model implementations. Average response time increased by 1.2 seconds (from 6.7s baseline to 7.9s with dual-LLM), primarily due to sequential processing through LLM-1 validation. Breakdown analysis shows: LLM-1 security checking contributes 0.9s (75% of overhead), inter-component communication adds 0.2s, and Response Handler validation adds 0.1s. During peak load (100+ concurrent users), latency increased to 9.1s average, remaining within acceptable parameters for educational applications. Regarding computational costs, the dual-LLM approach increases API calls by 100% (2 calls minimum per query), resulting in approximately 1.8x cost increase when accounting for cached responses and early rejections. However, this cost is offset by preventing security breaches—a single successful attack could result in reputational damage and remediation costs far exceeding the operational overhead.

The system throughput analysis demonstrates that the architecture maintains scalability characteristics that are appropriate for institutional deployment. Under peak load conditions simulating the maximum registration period traffic, the system sustained 95% of the baseline throughput, while maintaining complete security coverage. Load-balancing optimization and caching strategies further minimize the performance impact under normal operational conditions.

The dual-LLM architecture incorporates adaptive learning mechanisms that enable continuous improvements in response to evolve threat landscapes. The comprehensive logging infrastructure captures detailed attack attempt patterns, enabling machine-learning-based enhancement of detection capabilities through iterative model refinement. Administrative review processes for borderline cases provide valuable training data for improving classification accuracy and reducing false-positive rates, ensuring that the security model evolves in alignment with institutional policies and educational objectives.

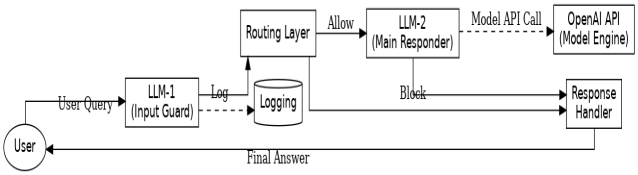


FIGURE 8. Conceptual flowchart of the moderation process.

The architecture supports integration with external threat intelligence sources, enabling proactive defence against newly identified attack patterns and emerging adversarial methodologies. This capability ensures that the ChatTEDU system remains protected against threats that extend beyond the initial observation period, while contributing to the broader cybersecurity community's understanding of open-access AI system vulnerabilities. The modular design facilitates the rapid deployment of security updates and model enhancements without system downtime or user experience disruption, which is essential for maintaining security effectiveness in dynamic threat environments while preserving educational service availability.

A. SYSTEM PERFORMANCE

The operational deployment of the dual-LLM architecture during the evaluation period of July–August 2024 demonstrated exceptional performance across multiple security and functional metrics. Empirical testing validated the architectural design principles outlined in Section V, confirming the effectiveness of layered security approaches in protecting educational AI systems against sophisticated adversarial threats.

Quantitative analysis revealed that LLM-1 achieved perfect detection accuracy (100%) across all the identified attack categories and successfully intercepted 180 malicious inputs without a single false negative. This comprehensive detection capability included prompt injection attempts (n=54), jailbreak methodologies (n=86), offensive content submissions (n=28), nonsense flooding attacks (n=11), and spam insertion attempts (n=1). The detection system maintained this performance level throughout the varying operational conditions, including the peak usage periods that coincided with university registration activities.

It is important to note that the 180 malicious queries were identified and labeled during live deployment through real-time monitoring, not used for system training or tuning. Ground truth labeling was performed by two independent